printed at home, october 2026

12 essays, 18,175 words. all of them are at paulgraham.com, free, as they should be. this is one reader's copy, printed at home to be read away from a screen, with a pencil.

a printed book of essays
github.com/workinprogmess/a-printed-book-of-essays

01

writing, briefly

march 2005
paulgraham.com/writing44.html

02

write simply

march 2021
paulgraham.com/simply.html

03

write like you talk

october 2015
paulgraham.com/talk.html

would sound like them talking.

if you simply manage to write in spoken language, you'll be ahead of 95% of writers. and it's so easy to do: just don't let a sentence through unless it's the way you'd say it to a friend.

thanks to patrick collison and jessica livingston for reading drafts of this.

04

writes and write-nots

october 2024
paulgraham.com/writes.html

05

putting ideas into words

february 2022
paulgraham.com/words.html

putting ideas into words doesn't have to mean writing, of course. you can also do it the old way, by talking. but in my experience, writing is the stricter test. you have to commit to a single, optimal sequence of words. less can go unsaid when you don't have tone of voice to carry meaning. and you can focus in a way that would seem excessive in conversation. i'll often spend 2 weeks on an essay and reread drafts 50 times. if you did that in conversation it would seem evidence of some kind of mental disorder. if you're lazy, of course, writing and talking are equally useless. but if you want to push yourself to get things right, writing is the steeper hill. [3]

the reason i've spent so long establishing this rather obvious point is that it leads to another that many people will find shocking. if writing down your ideas always makes them more precise and more complete, then no one who hasn't written about a topic has fully formed ideas about it. and someone who never writes has no fully formed ideas about anything nontrivial.

it feels to them as if they do, especially if they're not in the habit of critically examining their own thinking. ideas can feel complete. it's only when you try to put them into words that you discover they're not. so if you never subject your ideas to that test, you'll not only never have fully formed ideas, but also never realize it.

putting ideas into words is certainly no guarantee that they'll be right. far from it. but though it's not a sufficient condition, it is a necessary one.

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notes

1 machinery and circuits are formal languages.

2 i thought of this sentence as i was walking down the street in palo alto.

3 there are two senses of talking to someone: a strict sense in which the conversation is verbal, and a more general sense in which it can take any form, including writing. in the limit case (e.g. seneca's letters), conversation in the latter sense becomes essay writing.

it can be very useful to talk (in either sense) with other people as you're writing something. but a verbal conversation will never be more exacting than when you're talking about something you're writing.

thanks to trevor blackwell, patrick collison, and robert morris for reading drafts of this.

06

how to write usefully

february 2020
paulgraham.com/useful.html

my strategy is loose, then tight. i write the first draft of an essay fast, trying out all kinds of ideas. then i spend days rewriting it very carefully.

i've never tried to count how many times i proofread essays, but i'm sure there are sentences i've read 100 times before publishing them. when i proofread an essay, there are usually passages that stick out in an annoying way, sometimes because they're clumsily written, and sometimes because i'm not sure they're true. the annoyance starts out unconscious, but after the tenth reading or so i'm saying "ugh, that part" each time i hit it. they become like briars that catch your sleeve as you walk past. usually i won't publish an essay till they're all gone — till i can read through the whole thing without the feeling of anything catching.

i'll sometimes let through a sentence that seems clumsy, if i can't think of a way to rephrase it, but i will never knowingly let through one that doesn't seem correct. you never have to. if a sentence doesn't seem right, all you have to do is ask why it doesn't, and you've usually got the replacement right there in your head.

this is where essayists have an advantage over journalists. you don't have a deadline. you can work for as long on an essay as you need to get it right. you don't have to publish the essay at all, if you can't get it right. mistakes seem to lose courage in the face of an enemy with unlimited resources. or that's what it feels like. what's really going on is that you have different expectations for yourself. you're like a parent saying to a child "we can sit here all night till you eat your vegetables." except you're the child too.

i'm not saying no mistake gets through. for example, i added condition (c) in "a way to detect bias" after readers pointed out that i'd omitted it. but in practice you can catch nearly all of them.

there's a trick for getting importance too. it's like the trick i suggest to young founders for getting startup ideas: to make something you yourself want. you can use yourself as a proxy for the reader. the reader is not completely unlike you, so if you write about topics that seem important to you, they'll probably seem important to a significant number of readers as well.

importance has two factors. it's the number of people something matters to, times how much it matters to them. which means of course that it's not a rectangle, but a sort of ragged comb, like a riemann sum.

the way to get novelty is to write about topics you've thought about a lot. then you can use yourself as a proxy for the reader in this department too. anything you notice that surprises you, who've thought about the topic a lot, will probably also surprise a significant number of readers. and here, as with correctness and importance, you can use the morris technique to ensure that you will. if you don't learn anything from writing an essay, don't publish it.

you need humility to measure novelty, because acknowledging the novelty of an idea means acknowledging your previous ignorance of it. confidence and humility are often seen as opposites, but in this case, as in many others, confidence helps you to be humble. if you know you're an expert on some topic, you can freely admit when you learn something you didn't know, because you can be confident that most other people wouldn't know it either.

the fourth component of useful writing, strength, comes from two things: thinking well, and the skillful use of qualification. these two counterbalance each other, like the accelerator and clutch in a car with a manual transmission. as you try to refine the expression of an idea, you adjust the qualification accordingly. something you're sure of, you can state baldly with no qualification at all, as i did the four components of useful writing. whereas points that seem dubious have to be held at arm's length with perhapses.

it can sometimes work to deliberately phrase statements more weakly than you mean. to put "perhaps" in front of something you're actually quite sure of. but you'll notice that when writers do this, they usually do it with a wink.

i don't like to do this too much. it's cheesy to adopt an ironic tone for a whole essay. i think we just have to face the fact that elegance and curtness are two names for the same thing.

you might think that if you work sufficiently hard to ensure that an essay is correct, it will be invulnerable to attack. that's sort of true. it will be invulnerable to valid attacks. but in practice that's little consolation.

in fact, the strength component of useful writing will make you particularly vulnerable to misrepresentation. if you've stated an idea as strongly as you could without making it false, all anyone has to do is to exaggerate slightly what you said, and now it is false.

much of the time they're not even doing it deliberately. one of the most surprising things you'll discover, if you start writing essays, is that people who disagree with you rarely disagree with what you've actually written. instead they make up something you said and disagree with that.

for what it's worth, the countermove is to ask someone who does this to quote a specific sentence or passage you wrote that they believe is false, and explain why. i say "for what it's worth" because they never do. so although it might seem that this could get a broken discussion back on track, the truth is that it was never on track in the first place.

should you explicitly forestall likely misinterpretations? yes, if they're misinterpretations a reasonably smart and well-intentioned person might make. in fact it's sometimes better to say something slightly misleading and then add the correction than to try to get an idea right in one shot. that can be more efficient, and can also model the way such an idea would be discovered.

but i don't think you should explicitly forestall intentional misinterpretations in the body of an essay. an essay is a place to meet honest readers. you don't want to spoil your house by putting bars on the windows to protect against dishonest ones. the place to protect against intentional misinterpretations is in end-notes. but don't think you can predict them all. people are as ingenious at misrepresenting you when you say something they don't want to hear as they are at coming up with rationalizations for things they want to do but know they shouldn't. i suspect it's the same skill.

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as with most other things, the way to get better at writing essays is to practice. but how do you start? now that we've examined the structure of useful writing, we can rephrase that question more precisely. which constraint do you relax initially? the answer is, the first component of importance: the number of people who care about what you write.

if you narrow the topic sufficiently, you can probably find something you're an expert on. write about that to start with. if you only have ten readers who care, that's fine. you're helping them, and you're writing. later you can expand the breadth of topics you write about.

the other constraint you can relax is a little surprising: publication. writing essays doesn't have to mean publishing them. that may seem strange now that the trend is to publish every random thought, but it worked for me. i wrote what amounted to essays in notebooks for about 15 years. i never published any of them and never expected to. i wrote them as a way of figuring things out. but when the web came along i'd had a lot of practice.

07

good writing

may 2025
paulgraham.com/goodwriting.html

of ideas, and also makes mistakes easier to see. but now that we've dissolved one layer of preposterousness, i can't resist adding another. does sounding good do more than just help you get the ideas right? is writing that sounds good *inherently* more likely to be right? crazy as it may seem, i think that's true too.

obviously there's a connection at the level of individual words. there are lots of words in english that sound like what they mean, often in wonderfully subtle ways. glitter. round. scrape. prim. cavalcade. but the sound of good writing depends even more on the way you put words together, and there's a connection at that level too.

when writing sounds good, it's mostly because it has good rhythm. but the rhythm of good writing is not the rhythm of music, or the meter of verse. it's not so regular. if it were, it wouldn't be good, because the rhythm of good writing has to match the ideas in it, and ideas have all kinds of different shapes. sometimes they're simple and you just state them. but other times they're more subtle, and you need longer, more complicated sentences to tease out all the implications.

an essay is a cleaned up train of thought, in the same way dialogue is cleaned up conversation, and a train of thought has a natural rhythm. so when an essay sounds good, it's not merely because it has a pleasing rhythm, but because it has its natural one. which means you can use getting the rhythm right as a heuristic for getting the ideas right. and not just in principle: good writers do both simultaneously as a matter of course. often i don't even distinguish between the two problems. i just think ugh, this doesn't sound right; what do i mean to say here? [3]

the sound of writing turns out to be more like the shape of a plane than the color of a car. if it looks good, as kelly johnson used to say, it will fly well.

this is only true of writing that's used to develop ideas, though. it doesn't apply when you have ideas in some other way and then write about them afterward — for example, if you build something, or conduct an experiment, and then write a paper about it. in such cases the ideas often live more in the work than the writing, so the writing can be bad even though the ideas are good. the writing in textbooks and popular surveys can be bad for the same reason: the author isn't developing the ideas, merely describing other people's. it's only when you're writing to develop ideas that there's such a close connection between the two senses of doing it well.

ok, many people will be thinking, this seems plausible so far, but what about liars? is it not notoriously possible for a smooth-tongued liar to write something beautiful that's completely false?

it is, of course. but not without method acting. the way to write something beautiful and false is to begin by making yourself almost believe it. so just like someone writing something beautiful and true, you're presenting a perfectly-formed train of thought. the difference is the point where it attaches to the world. you're saying something that would be true if certain false premises were. if for some bizarre reason the number of jobs in a country were fixed, then immigrants really would be taking our jobs.

so it's not quite right to say that better sounding writing is more likely to be true. better sounding writing is more likely to be internally consistent. if the writer is honest, internal consistency and truth converge.

but while we can't safely conclude that beautiful writing is true, it's usually safe to conclude the converse: something that seems clumsily written will usually have gotten the ideas wrong too.

08

the best essay

march 2024
paulgraham.com/best.html

if you come across a question that's sufficiently puzzling, it could be worth exploring even if it doesn't seem very momentous. many an important discovery has been made by pulling on a thread that seemed insignificant at first. how can they *all* be finches? [2]

once you've got a question, then what? you start thinking out loud about it. not literally out loud, but you commit to a specific string of words in response, as you would if you were talking. this initial response is usually mistaken or incomplete. writing converts your ideas from vague to bad. but that's a step forward, because once you can see the brokenness, you can fix it.

perhaps beginning writers are alarmed at the thought of starting with something mistaken or incomplete, but you shouldn't be, because this is why essay writing works. forcing yourself to commit to some specific string of words gives you a starting point, and if it's wrong, you'll see that when you reread it. at least half of essay writing is rereading what you've written and asking *is this correct and complete?* you have to be very strict when rereading, not just because you want to keep yourself honest, but because a gap between your response and the truth is often a sign of new ideas to be discovered.

the prize for being strict with what you've written is not just refinement. when you take a roughly correct answer and try to make it exactly right, sometimes you find that you can't, and that the reason is that you were depending on a false assumption. and when you discard it, the answer turns out to be completely different. [3]

ideally the response to a question is two things: the first step in a process that converges on the truth, and a source of additional questions (in my very general sense of the word). so the process continues recursively, as response spurs response. [4]

usually there are several possible responses to a question, which means you're traversing a tree. but essays are linear, not tree-shaped, which means you have to choose one branch to follow at each point. how do you choose? usually you should follow whichever offers the greatest combination of generality and novelty. i don't consciously rank branches this way; i just follow whichever seems most exciting; but generality and novelty are what make a branch exciting. [5]

if you're willing to do a lot of rewriting, you don't have to guess right. you can follow a branch and see how it turns out, and if it isn't good enough, cut it and backtrack. i do this all the time. in this essay i've already cut a 17-paragraph subtree, in addition to countless shorter ones. maybe i'll reattach it at the end, or boil it down to a footnote, or spin it off as its own essay; we'll see. [6]

in general you want to be quick to cut. one of the most dangerous temptations in writing (and in software and painting) is to keep something that isn't right, just because it contains a few good bits or cost you a lot of effort.

the most surprising new question being thrown off at this point is *does it really matter what the initial question is?* if the space of ideas is highly connected, it shouldn't, because you should be able to get from any question to the most valuable ones in a few hops. and we see evidence that it's highly connected in the way, for example, that people who are obsessed with some topic can turn any conversation toward it. but that only works if you know where you want to go, and you don't in an essay. that's the whole point. you don't want to be the obsessive conversationalist, or all your essays will be about the same thing. [7]

the other reason the initial question matters is that you usually feel somewhat obliged to stick to it. i don't think about this when i decide which branch to follow. i just follow novelty and generality. sticking to the question is enforced later, when i notice i've wandered too far and have to backtrack. but i

you can probably cause yourself to have more of them, though. the quality of the ideas that come out of your head depends on what goes in, and you can improve that in two dimensions, breadth and depth.

you can't learn everything, so getting breadth implies learning about topics that are very different from one another. when i tell people about my book-buying trips to hay and they ask what i buy books about, i usually feel a bit sheepish answering, because the topics seem like a laundry list of unrelated subjects. but perhaps that's actually optimal in this business.

you can also get ideas by talking to people, by doing and building things, and by going places and seeing things. i don't think it's important to talk to new people so much as the sort of people who make you have new ideas. i get more new ideas after talking for an afternoon with robert morris than from talking to 20 new smart people. i know because that's what a block of office hours at y combinator consists of.

while breadth comes from reading and talking and seeing, depth comes from doing. the way to really learn about some domain is to have to solve problems in it. though this could take the form of writing, i suspect that to be a good essayist you also have to do, or have done, some other kind of work. that may not be true for most other fields, but essay writing is different. you could spend half your time working on something else and be net ahead, so long as it was hard.

i'm not proposing that as a recipe so much as an encouragement to those already doing it. if you've spent all your life so far working on other things, you're already halfway there. though of course to be good at writing you have to like it, and if you like writing you'd probably have spent at least some time doing it.

everything i've said about initial questions applies also to the questions you encounter in writing the essay. they're the same thing; every subtree of an essay is usually a shorter essay, just as every subtree of a calder mobile is a smaller mobile. so any technique that gets you good initial questions also gets you good whole essays.

at some point the cycle of question and response reaches what feels like a natural end. which is a little suspicious; shouldn't every answer suggest more questions? i think what happens is that you start to feel sated. once you've covered enough interesting ground, you start to lose your appetite for new questions. which is just as well, because the reader is probably feeling sated too. and it's not lazy to stop asking questions, because you could instead be asking the initial question of a new essay.

that's the ultimate source of drag on the connectedness of ideas: the discoveries you make along the way. if you discover enough starting from question a, you'll never make it to question b. though if you keep writing essays you'll gradually fix this problem by burning off such discoveries. so bizarrely enough, writing lots of essays makes it as if the space of ideas were more highly connected.

when a subtree comes to an end, you can do one of two things. you can either stop, or pull the cubist trick of laying separate subtrees end to end by returning to a question you skipped earlier. usually it requires some sleight of hand to make the essay flow continuously at this point, but not this time. this time i actually need an example of the phenomenon. for example, we discovered earlier that the best possible essay wouldn't usually be timeless in the way the best painting would. this seems surprising enough to be worth investigating further.

there are two senses in which an essay can be timeless: to be about a matter of permanent importance, and always to have the same effect on readers. with art these two senses blend together. art that looked beautiful to the ancient greeks still looks beautiful to us. but with essays the two senses diverge,

i already aim for it. breadth and novelty are the two things i'm always chasing. but i'm glad i understand where timelessness fits.

i understand better where a lot of things fit now. this essay has been a kind of tour of essay writing. i started out hoping to get advice about topics; if you assume good writing, the only thing left to differentiate the best essay is its topic. and i did get advice about topics: discover natural selection. yeah, that would be nice. but when you step back and ask what's the best you can do short of making some great discovery like that, the answer turns out to be about procedure. ultimately the quality of an essay is a function of the ideas discovered in it, and the way you get them is by casting a wide net for questions and then being very exacting with the answers.

the most striking feature of this map of essay writing are the alternating stripes of inspiration and effort required. the questions depend on inspiration, but the answers can be got by sheer persistence. you don't have to get an answer right the first time, but there's no excuse for not getting it right eventually, because you can keep rewriting till you do. and this is not just a theoretical possibility. it's a pretty accurate description of the way i work. i'm rewriting as we speak.

but although i wish i could say that writing great essays depends mostly on effort, in the limit case it's inspiration that makes the difference. in the limit case, the questions are the harder thing to get. that pool has no bottom.

how to get more questions? that is the most important question of all.

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notes

1 there might be some resistance to this conclusion on the grounds that some of these discoveries could only be understood by a small number of readers. but you get into all sorts of difficulties if you want to disqualify essays on this account. how do you decide where the cutoff should be? if a virus kills off everyone except a handful of people sequestered at los alamos, could an essay that had been disqualified now be eligible? etc.

darwin's 1844 essay was derived from an earlier version written in 1839. extracts from it were published in 1858.

2 when you find yourself very curious about an apparently minor question, that's an exciting sign. evolution has designed you to pay attention to things that matter. so when you're very curious about something random, that could mean you've unconsciously noticed it's less random than it seems.

3 corollary: if you're not intellectually honest, your writing won't just be biased, but also boring, because you'll miss all the ideas you'd have discovered if you pushed for the truth.

4 sometimes this process begins before you start writing. sometimes you've already figured out the first few things you want to say. schoolchildren are often taught they should decide *everything* they want to say, and write this down as an outline before they start writing the essay itself. maybe that's a good way to get them started — or not, i don't know — but it's antithetical to the spirit of essay writing. the more detailed your outline, the less your ideas can benefit from the sort of discovery that essays are for.

5 the problem with this type of "greedy" algorithm is that you can end up on a local maximum. if the most valuable question is preceded by a boring one, you'll overlook it. but i can't imagine a better strategy. there's no lookahead except by writing. so use a greedy algorithm and a lot of time.

6 i ended up reattaching the first 5 of the 17 paragraphs, and discarding the rest.

7 stephen fry confessed to making use of this phenomenon when taking exams at oxford. he had in his head a standard essay about some general literary topic, and he would find a way to turn the exam question toward

09

the shape of the essay field

june 2025
paulgraham.com/field.html

thinking. what would surprise young readers? which important things do people tend to learn late? interesting question. i should think about that.

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notes

1 it's hard to write a really good essay about an unimportant topic, though, because a really good essayist will inevitably draw the topic into deeper waters. e. b. white could write an essay about how to boil potatoes that ended up being full of timeless wisdom. in which case, of course, it wouldn't really be about how to boil potatoes; that would just have been the starting point.

thanks to jessica livingston and michael nielsen for reading drafts of this.

10

the age of the essay

september 2004
paulgraham.com/essay.html

essors who taught history could be required to write scholarly articles about history, but what about the professors who taught rhetoric or composition? what should they do research on? the closest thing seemed to be english literature.[3]

and so in the late 19th century the teaching of writing was inherited by english professors. this had two drawbacks: (a) an expert on literature need not himself be a good writer, any more than an art historian has to be a good painter, and (b) the subject of writing now tends to be literature, since that's what the professor is interested in.

high schools imitate universities. the seeds of our miserable high school experiences were sown in 1892, when the national education association "formally recommended that literature and composition be unified in the high school course."[4] the 'riting component of the 3 rs then morphed into english, with the bizarre consequence that high school students now had to write about english literature-- to write, without even realizing it, imitations of whatever english professors had been publishing in their journals a few decades before.

it's no wonder if this seems to the student a pointless exercise, because we're now three steps removed from real work: the students are imitating english professors, who are imitating classical scholars, who are merely the inheritors of a tradition growing out of what was, 700 years ago, fascinating and urgently needed work.

no defense

the other big difference between a real essay and the things they make you write in school is that a real essay doesn't take a position and then defend it. that principle, like the idea that we ought to be writing about literature, turns out to be another intellectual hangover of long forgotten origins.

it's often mistakenly believed that medieval universities were mostly seminaries. in fact they were more law schools. and at least in our tradition lawyers are advocates, trained to take either side of an argument and make as good a case for it as they can. whether cause or effect, this spirit pervaded early universities. the study of rhetoric, the art of arguing persuasively, was a third of the undergraduate curriculum.[5] and after the lecture the most common form of discussion was the disputation. this is at least nominally preserved in our present-day thesis defense: most people treat the words thesis and dissertation as interchangeable, but originally, at least, a thesis was a position one took and the dissertation was the argument by which one defended it.

defending a position may be a necessary evil in a legal dispute, but it's not the best way to get at the truth, as i think lawyers would be the first to admit. it's not just that you miss subtleties this way. the real problem is that you can't change the question.

and yet this principle is built into the very structure of the things they teach you to write in high school. the topic sentence is your thesis, chosen in advance, the supporting paragraphs the blows you strike in the conflict, and the conclusion-- uh, what is the conclusion? i was never sure about that in high school. it seemed as if we were just supposed to restate what we said in the first paragraph, but in different enough words that no one could tell. why bother? but when you understand the origins of this sort of "essay," you can see where the conclusion comes from. it's the concluding remarks to the jury.

good writing should be convincing, certainly, but it should be convincing because you got the right answers, not because you did a good job of arguing. when i give a draft of an essay to friends, there are two things i want to know: which parts bore them, and which seem unconvincing. the boring bits can

but *what* you tell him doesn't matter, so long as it's interesting. i'm sometimes accused of meandering. in defend-a-position writing that would be a flaw. there you're not concerned with truth. you already know where you're going, and you want to go straight there, blustering through obstacles, and hand-waving your way across swampy ground. but that's not what you're trying to do in an essay. an essay is supposed to be a search for truth. it would be suspicious if it didn't meander.

the meander (aka menderes) is a river in turkey. as you might expect, it winds all over the place. but it doesn't do this out of frivolity. the path it has discovered is the most economical route to the sea.[6]

the river's algorithm is simple. at each step, flow down. for the essayist this translates to: flow interesting. of all the places to go next, choose the most interesting. one can't have quite as little foresight as a river. i always know generally what i want to write about. but not the specific conclusions i want to reach; from paragraph to paragraph i let the ideas take their course.

this doesn't always work. sometimes, like a river, one runs up against a wall. then i do the same thing the river does: backtrack. at one point in this essay i found that after following a certain thread i ran out of ideas. i had to go back seven paragraphs and start over in another direction.

fundamentally an essay is a train of thought-- but a cleaned-up train of thought, as dialogue is cleaned-up conversation. real thought, like real conversation, is full of false starts. it would be exhausting to read. you need to cut and fill to emphasize the central thread, like an illustrator inking over a pencil drawing. but don't change so much that you lose the spontaneity of the original.

err on the side of the river. an essay is not a reference work. it's not something you read looking for a specific answer, and feel cheated if you don't find it. i'd much rather read an essay that went off in an unexpected but interesting direction than one that plodded dutifully along a prescribed course.

surprise

so what's interesting? for me, interesting means surprise. interfaces, as geoffrey james has said, should follow the principle of least astonishment. a button that looks like it will make a machine stop should make it stop, not speed up. essays should do the opposite. essays should aim for maximum surprise.

i was afraid of flying for a long time and could only travel vicariously. when friends came back from faraway places, it wasn't just out of politeness that i asked what they saw. i really wanted to know. and i found the best way to get information out of them was to ask what surprised them. how was the place different from what they expected? this is an extremely useful question. you can ask it of the most unobservant people, and it will extract information they didn't even know they were recording.

surprises are things that you not only didn't know, but that contradict things you thought you knew. and so they're the most valuable sort of fact you can get. they're like a food that's not merely healthy, but counteracts the unhealthy effects of things you've already eaten.

how do you find surprises? well, therein lies half the work of essay writing. (the other half is expressing yourself well.) the trick is to use yourself as a proxy for the reader. you should only write about things you've thought about a lot. and anything you come across that surprises you, who've thought about the topic a lot, will probably surprise most readers.

for example, in a recent essay i pointed out that because you can only judge computer programmers by working with them, no one knows who the best programmers are overall. i didn't realize this when i began that essay, and even now i find it kind of weird. that's what you're looking for.

whole process seemed pointless. but none of us had the balls at the time to hypothesize that it was, in fact, all a mistake. we all thought there was just something we weren't getting.

i have a hunch you want to pay attention not just to things that seem wrong, but things that seem wrong in a humorous way. i'm always pleased when i see someone laugh as they read a draft of an essay. but why should i be? i'm aiming for good ideas. why should good ideas be funny? the connection may be surprise. surprises make us laugh, and surprises are what one wants to deliver.

i write down things that surprise me in notebooks. i never actually get around to reading them and using what i've written, but i do tend to reproduce the same thoughts later. so the main value of notebooks may be what writing things down leaves in your head.

people trying to be cool will find themselves at a disadvantage when collecting surprises. to be surprised is to be mistaken. and the essence of cool, as any fourteen year old could tell you, is *nil admirari.* when you're mistaken, don't dwell on it; just act like nothing's wrong and maybe no one will notice.

one of the keys to coolness is to avoid situations where inexperience may make you look foolish. if you want to find surprises you should do the opposite. study lots of different things, because some of the most interesting surprises are unexpected connections between different fields. for example, jam, bacon, pickles, and cheese, which are among the most pleasing of foods, were all originally intended as methods of preservation. and so were books and paintings.

whatever you study, include history-- but social and economic history, not political history. history seems to me so important that it's misleading to treat it as a mere field of study. another way to describe it is *all the data we have so far.*

among other things, studying history gives one confidence that there are good ideas waiting to be discovered right under our noses. swords evolved during the bronze age out of daggers, which (like their flint predecessors) had a hilt separate from the blade. because swords are longer the hilts kept breaking off. but it took five hundred years before someone thought of casting hilt and blade as one piece.

disobedience

above all, make a habit of paying attention to things you're not supposed to, either because they're "inappropriate," or not important, or not what you're supposed to be working on. if you're curious about something, trust your instincts. follow the threads that attract your attention. if there's something you're really interested in, you'll find they have an uncanny way of leading back to it anyway, just as the conversation of people who are especially proud of something always tends to lead back to it.

for example, i've always been fascinated by comb-overs, especially the extreme sort that make a man look as if he's wearing a beret made of his own hair. surely this is a lowly sort of thing to be interested in-- the sort of superficial quizzing best left to teenage girls. and yet there is something underneath. the key question, i realized, is how does the comber-over not see how odd he looks? and the answer is that he got to look that way *incrementally.* what began as combing his hair a little carefully over a thin patch has gradually, over 20 years, grown into a monstrosity. gradualness is very powerful. and that power can be used for constructive purposes too: just as you can trick yourself into looking like a freak, you can trick yourself into creating something so grand that you would never have dared to *plan* such a thing. indeed, this is just how most good software gets created. you start by writing a stripped-down kernel (how hard can it be?) and gradually it grows into a complete operating system. hence the next leap: could you do the same thing in painting, or in a novel?

6 trevor blackwell points out that this isn't strictly true, because the outside edges of curves erode faster.

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what you (want to)* want

november 2022
paulgraham.com/want.html

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how to get new ideas

january 2023
paulgraham.com/getideas.html